mathematical proof that, when a static star with a tiny mountain is made more and more compact, there are just two possible outcomes. Either the star creates a precisely spherical hole around itself, or else the mountain produces such enormous spacetime curvature, as the star nears its critical circumference, that the mountain's effects are no longer a "small perturbation"; the method of calculation then fails, and the outcome of the implosion is unknown. The second piece of the analysis was what I soon learned to identify as a "typical Zel'dovich" argument: If the mountain initially is tiny, it is *intuitively obvious* that the mountain *cannot* produce enormous curvature as the star nears its critical circumference. We must discard that possibility. The other possibility must be the truth: The star must produce a precisely spherical hole.

What was intuitively obvious to Zel'dovich (and would ultimately turn out to be true) was far from obvious to most Western physicists. Controversy began to swirl.

The power of a controversial research result is enormous. It attracts physicists like picnics attract ants. Thus it was with the Zel'dovich team's no-hair evidence. The physicists, like ants, came one by one at first, but then in droves.

The first was Werner Israel, born in Berlin, reared in South Africa, trained in the laws of relativity in Ireland, and now struggling to start a relativity research group in Edmonton, Canada. In a mathematical tour de force, Israel improved on the first, Doroshkevich–Novikov, part of the Soviet proof: He treated not just tiny mountains, as had the Soviets, but mountains of any size and shape. In fact, his calculations worked correctly for *any* implosion, no matter how nonspherical, even a square one, and they allowed the implosion to be dynamical, not just an idealized sequence of static stars. Equally remarkable was Israel's conclusion, which was similar to the Doroshkevich–Novikov conclusion, but far stronger: *A highly nonspherical implosion can have only two outcomes: either it produces no black hole at all, or else it produces a black hole that is precisely spherical* For this conclusion to be true, however, the imploding body had to have two special properties: It must be completely devoid of any electric charge, and it must not spin at all. The reasons will become clear below.

Israel first presented his analysis and results on 8 February 1967, at a lecture at Kings College in London. The title of the lecture was a little